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Fenestration system with head slide

Abstract

A sliding and pivoting fenestration unit, such as a sliding door, with screen. Embodiments of the fenestration unit include a locking and height-adjustable pivot rider, a head slide, a screen carrier, and a method of operation.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) Reference is hereby made to the following applications filed on even date herewith: 1. application Ser. No. 18/603,320 entitled Pivoting Fenestration System with Pivot Wheel Assembly. 2. application Ser. No. 18/603,330 entitled Locking and Height-Adjustable Pivot Rider. 3. application Ser. No. 18/603,345 entitled Fenestration System with Screen Carrier.

TECHNICAL FIELD

(2) Various aspects of this disclosure relate to fenestration products, such as sliding doors and windows. In some specific examples, the disclosure relates to sliding doors and windows with one or more hinged or pivoting panels, and which may also include a screen.

BACKGROUND

(3) Sliding doors and windows also capable of hinged or pivotal motion to open positions outside of their frames are generally known, and disclosed, for example, in U.S. Patent Application Publication 2020/0217124 entitled Sliding and Pivot Fenestration Unit, which is incorporated herein by reference in its entirety and for all purposes.

(4) There remains a continuing need for improved sliding and pivot fenestration units and components for such units.

SUMMARY

(5) Various examples of this disclosure relate to fenestration units and systems, such as for example doors and windows, with first and/or second panels mounted with respect to a frame. Some aspects relate to slide and/or pivot capabilities and features of the units and panels. Aspects also relate to components to provide the slide and/or pivot features and/or capabilities of the panels. Yet other aspects relate to methods for sliding and/or pivoting operation of the panels.

(6) A first example is a fenestration unit with a locking pivot wheel assembly. The fenestration unit may comprise a frame, a first panel supported by the frame, and a locking pivot rider assembly. In embodiments, the frame defines a frame plane and includes a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb. The locking pivot rider assembly couples the first panel to the frame and facilitates slide mode operation and pivot mode operation of the first panel. During the slide mode operation, the first panel is in a slide position generally parallel to the frame plane and can slide within the frame between the first and second jambs. During the pivot mode operation the first panel can pivot outwardly to a pivot open position beyond the frame plane. The locking pivot rider assembly may include a rider coupling the first panel to the frame for the slide mode operation; a pivot rotatably coupling the first panel to the rider for the pivot mode operation; and a lock coupled to the rider and the pivot, to releasably lock the rider with respect to the frame. Rotation of the pivot during the pivot mode operation of the first panel from the slide position to the pivot open position actuates the lock and causes the lock to inhibit sliding movement of the rider on the frame. Rotation of the pivot during the pivot mode operation of the first panel from the pivot open position to the slide position releases the lock and allows sliding movement of the rider on the frame.

(7) In some embodiments, the lock is configured to engage the rider to the frame when actuated, and to disengage the rider from the frame when released.

(8) In any or all of these embodiments of this first example, the lock may include one or more lock pins, each lock pin movable within the rider between a retracted position disengaged from the frame, and an extended position engaged to the frame; and an actuator coupling the pivot to each of the one or more lock pins, wherein the actuator causes each of the lock pins to move from the retracted position to the extended position when the first panel rotates from the slide position to the pivot open position.

(9) In any or all of these embodiments of this first example, the lock may further include one or more biasing members for biasing each of the one or more lock pins from the extended position to the retracted position.

- (10) In any or all of these embodiments of the first example, the actuator may include a cam coupling the pivot to each of the one or more lock pins, wherein the cam rotates with the pivot.
- (11) In any or all of these embodiments of the first example, the rider may include a first end which for example is shown in the illustrated embodiments as a conical surface including a central apex region; a second end opposite the first end; one or more bores extending between and through the conical or other surface of the first end and the second end, and wherein each of the one or more bores receives one of the one or more lock pins; and wherein the cam rotates on the first end of the rider. Other embodiments of the rider have a first end with other (e.g., not conical) surfaces (not shown)
- (12) Any or all of these embodiments of the first example may further include a releasable lock to inhibit sliding motion of the first panel during an initial portion of the rotation of the first panel from the pivot ready position to the pivot open position before at least one of the one or more lock pins is moved to the extended position, and to inhibit sliding motion of the first panel during a final portion of the rotation of the first panel from the pivot open position to the pivot ready position after all of the one or more lock pins is moved from the retracted position. In embodiments, the releasable lock may include a panel member mounted to the first panel; and a sill member mounted to the sill, wherein the panel member and the sill member engage to facilitate rotation of the first panel, and to inhibit sliding motion of the first panel, during the initial portion and the final portion of the rotation of the first panel.
- (13) Any or all of these embodiments of the first example may further include a height adjuster to adjust a height of the first panel with respect to the rider. The height adjuster may include a height-adjustable mount coupling the pivot to the rider; and an adjustment mechanism for actuating the height-adjustable mount. In some embodiments, the height-adjustable mount includes a threaded shaft coupling the pivot to the rider; and the adjustment mechanism includes a rotating mechanism to rotate the threaded shaft. In some embodiments, the adjustment mechanism is accessible and operable from an exterior of the first panel.
- (14) In any or all of these embodiments of the first example, the adjustment mechanism may include a beveled gear set.
- (15) In any or all of these embodiments of the first example, the pivot may include a bearing plate mounted to the first panel, and wherein the threaded shaft is rotatably coupled to the bearing plate.
- (16) In any or all of these embodiments of the first example, the rider may include a housing; and a wheel rotatably mounted to the housing and positioned on the frame.
- (17) In any or all of these embodiments of the first example, the locking pivot rider assembly may couple a bottom portion of the first panel to the sill of the frame.
- (18) In any or all of these embodiments of the first example, when in the slide mode the first panel may be configured to slide between a slide closed position, a slide fully open position, and a pivot ready position, and wherein the pivot ready position is beyond the slide fully open position from the slide closed position.
- (19) A second example is a fenestration unit including a second panel in accordance with any or all of the above embodiments of the first example. The fenestration unit of the second example may further comprise a second panel supported by the frame; a hinge coupling the second panel to the frame and to facilitate hinge mode operation of the second panel, wherein during the hinge mode operation the second panel can pivot between a closed position generally parallel to the frame plane and an open position beyond the frame plane; wherein when the first panel is in the slide closed position and the second panel is in the closed position, the first panel and the second panel are parallel with one another, and wherein when the first panel is in the slide fully open position and the second panel is in the closed position, the first and second panels are parallel with one another and the first panel is at a lateral position that at least partially overlaps the second panel; and one or more stop members configured to prevent the first panel from sliding beyond its slide fully open position toward the pivot ready position during the slide mode operation when the second panel is

in its closed position.

(20) In any or all embodiments of the second example, each of the stop members may comprise a bumper coupled to the second panel. In embodiments, the stop members may be flexible members.

(21) In any or all embodiments of the second example, the lock and/or frame may be configured to cause the lock to be disengaged with respect to the frame when the first panel is between its slide closed and slide fully open positions, and to allow the lock to engage with respect to the frame when the first panel is in the pivot ready position. The fenestration unit may further comprise a rider on the first panel at a location spaced apart from the locking pivot rider assembly; a track on the frame, wherein the rider is configured to engage the track during the slide mode operation of the first panel between its slide closed and slide fully open positions; and a gap in the track, wherein the gap is located at a position corresponding to the position of the rider when the first panel is at the pivot ready position, thereby allowing the rider to disengage from the track during the pivot mode operation of the first panel.

(22) A third example is a locking pivot rider assembly. The rider assembly of the third example may support a panel on a frame, and may comprise a rider, a pivot, and a lock. The rider is configured to slide on the frame. The pivot is configured to rotatably couple the rider to the panel. The pivot is configured to rotate between a first position and a second position with rotation of the panel between a pivot closed position and a pivot open position with respect to the frame. The lock is coupled to the rider and the pivot to releasably lock the rider with respect to the frame. Rotation of the pivot from the first position to the second position in response to rotation of the panel from the pivot closed position toward the pivot open position actuates the lock to inhibit the rider from sliding on the frame. Rotation of the pivot from the second position to the first position in response to rotation of the panel from the pivot open position to the pivot closed position causes the lock to release and allow the rider to slide on the frame.

(23) In some embodiments of the third example, the lock may be configured to engage the rider to the frame when actuated, and to disengage the rider from the frame when released.

(24) In any or all embodiments of the third example, the lock may include one or more lock pins, each lock pin movable within the rider between a retracted position disengaged from the frame, and an extended position engaged to the frame; and an actuator coupling the pivot to each of the one or more lock pins, wherein the actuator causes each of the lock pins to move from the retracted position to the extended position when the first panel rotates from the slide position to the pivot open position. In embodiments, the lock may further include one or more biasing members for biasing each of the one or more lock pins from the extended position to the retracted position. In embodiments, the actuator may include a cam coupling the pivot to each of the one or more lock pins, wherein the cam rotates with the pivot. In embodiments, the rider may include a first end having a conical surface including a central apex region; a second end opposite the first end; one or more bores extending between and through the conical surface of the first end and the second end, and wherein each of the one or more bores receives one of the one or more lock pins; and wherein the cam rotates on the first end of the rider.

(25) Any or all embodiments of the third example may further include a height adjuster to adjust a height of the pivot with respect to the rider. In embodiments, the height adjuster may include a height-adjustable mount coupling the pivot the rider; and an adjustment mechanism for actuating the height-adjustable mount. In some embodiments, the height-adjustable mount includes a threaded shaft coupling the pivot to the rider; and the adjustment mechanism includes a rotating mechanism to rotate the threaded shaft.

(26) In any or all embodiments of the third example, the adjustment mechanism may be accessible and operable from an exterior of the panel.

(27) In any or all embodiments of the third example, the adjustment mechanism includes a beveled gear set.

(28) In any or all embodiments of the third example, the pivot may include a bearing plate mounted

to the panel, and wherein the threaded shaft is rotatably coupled to the bearing plate.

(29) In any or all embodiments of the third example, the rider may include a housing; and a wheel rotatably mounted to the housing.

(30) A fourth example is a height-adjustable rider assembly. The rider assembly of the fourth example may support a panel on a frame, and may comprise a rider, a bearing plate, a height-adjustable mount, and an adjustment mechanism. The rider may be configured to slide on the frame. The bearing plate may be configured to be coupled to the panel. The height-adjustable mount may couple the rider to the bearing plate. The adjustment mechanism actuates the height-adjustable mount, to raise and lower the bearing plate with respect to the rider.

(31) In some embodiments of the fourth example, the height-adjustable mount may include a threaded shaft coupling the bearing plate to the rider; and the adjustment mechanism may include a rotating mechanism to rotate the threaded shaft.

(32) In any or all embodiments of the fourth example, the adjustment mechanism may be accessible and operable from an exterior of the panel.

(33) In any or all embodiments of the fourth example, the adjustment mechanism may include a beveled gear set configured to rotate the threaded shaft.

(34) In any or all embodiments of the fourth example, the panel may include an opening providing access the adjustment mechanism.

(35) In any or all embodiments of the fourth example, the rider may include a housing; and a wheel rotatably mounted to the housing.

(36) A fifth example is a fenestration system with a head slide. Embodiments of the fifth example may include a frame, a first panel, a slide assembly, and a latch actuator. The frame may include a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb. The first panel may be supported by the frame. The slide assembly couples the first panel to the frame to facilitate slide mode operation and pivot mode operation of the first panel. During the slide mode operation the first panel can slide within the frame between the first and second jambs when the first panel is in a slide position generally parallel to the frame plane. During the pivot mode operation the first panel can pivot outwardly beyond the frame plane. Embodiments of the slide assembly include a slide body configured for sliding motion between the first and second jambs during the slide mode operation; a pivot rotatably coupling a second end portion of the first panel to a second end portion of the slide body; and a latch releasably coupling a first end portion of the first panel to a first end portion of the slide body, and operable between a latched position and an unlatched position. When the latch is in the latched position the latch couples the first panel to the slide body to facilitate the slide mode operation of the first panel. When the latch is in the unlatched position the latch releases the first panel from the slide body to facilitate the pivot mode operation of the first panel. The latch actuator may be coupled to the latch and operable by a user to cause the latch to move between the latched position and the unlatched position.

(37) In any or all embodiments of the fifth example, the latch actuator may be located on the first panel.

(38) In any or all embodiments of the fifth example, the slide assembly may be configured to slide the first panel between a slide closed position, a slide open position, and a pivot ready position beyond the slide open position from the slide closed position during the slide mode operation. The latch may further include a stop that inhibits actuation of the latch to the unlatched position unless the first panel is at the pivot ready position. In embodiments, the frame includes a slide assembly guide track, and the slide body of the slide assembly is slidably mounted in the slide assembly guide track. The slide assembly guide track may include a structure configured to cooperate with the stop of the latch when the first panel is at the pivot ready position to enable the actuation of the latch to the unlatched position. In embodiments, the guide track includes a structure configured to cooperate with the stop of the latch when the first panel is at and between its slide closed position and its slide open position, to inhibit the actuation of the latch to the unlatched position.

(39) In any or all embodiments of the fifth example, the frame may include a slide assembly guide track, and the slide body of the slide assembly may be slidably mounted in the slide assembly guide track. In embodiments, the slide assembly guide track is on the head of the frame, and the slide assembly couples a top portion of the first panel to the head of the frame.

(40) In any or all embodiments of the fifth example, the first panel is defined by first and second opposite ends; and the slide body may extend an entire distance between the first and second opposite ends of the first panel.

(41) In any or all embodiments of the fifth example, the latch may comprise a catch movably mounted to the first panel; and a pin that is actuated by the latch actuator to releasably engage the catch. In embodiments, movement of the pin causes the latch to move between the latched position and the unlatched position. In embodiments, the catch comprises a base member; and an engagement member extending from the base member, wherein the engagement member includes a slot that can be engaged and disengaged by the pin. In embodiments, the base member is pivotally mounted to the first panel for rotational movement about a pivot axis extending in a direction between a top portion and a bottom portion of the first panel; and the engagement member extends in a generally perpendicular direction from the base member. In embodiments, the base member is pivotally mounted to one of a top portion or a bottom portion of the first panel; and the engagement member extends in a direction generally parallel to an end edge of the first panel.

(42) Embodiments may further include a stop on the base member that inhibits actuation of the latch to the unlatched position unless the first panel is at the pivot ready position. In embodiments, the frame may include a slide assembly guide track, and wherein the slide body of the slide assembly is slidably mounted in the slide assembly guide track; and the slide assembly guide track includes a structure configured to cooperate with the stop on the base member when the first panel is at the pivot ready position to enable the actuation of the latch to the unlatched position. In embodiments, the guide track may include a structure configured to cooperate with the stop on the base member when the first panel is at and between its slide closed position and its slide open position, to inhibit the actuation of the latch to the unlatched position. In embodiments, the slot of the engagement member slopes to inhibit the actuation of the latch to the unlatched position when the first panel is at and between its slide closed and slide open positions.

(43) A sixth example is a fenestration system with a screen carrier. Embodiments of the fenestration system of the sixth example may comprise a frame defining a frame plane and including a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb; a second panel supported by the frame; a first panel supported by the frame; a screen supported by the frame; and screen carrier coupling the screen to the frame. The second panel may be configured for pivot mode operation, wherein during the pivot mode operation the second panel can pivot between a pivot closed position generally parallel to the frame plane and a pivot open position beyond the frame plane. The second panel extends over a second lateral portion of the frame when in its pivot closed position. The first panel may be configured for slide mode operation and pivot mode operation with respect to the frame. During the slide mode operation the first panel is in a slide position generally parallel to the frame plane and can slide within the frame between slide closed and slide open positions. During the pivot mode operation the first panel can pivot outwardly beyond the frame plane. When the first panel is in the slide closed position the first panel extends over a first lateral portion of the frame. The screen carrier may be configured to facilitate slide mode operation and pivot mode operation of the screen. During the slide mode operation the screen can slide within the frame between a slide closed position corresponding to the slide closed position of the first panel, a slide open position at least partially overlapping the second panel when the second panel is in its pivot closed position, and a pivot ready position engaged with the second panel when the second panel is in its pivot closed position. During its pivot mode operation the screen is in its pivot ready position and can move outwardly beyond the frame plane with the second panel during the pivot mode operation of the second panel.

(44) In embodiments of the fenestration system of the sixth example, the screen carrier may include a first section mounted to the frame; and a second section mounted to the second panel, wherein the screen transitions between the first and second sections during its slide mode operation. In embodiments, the first section of the screen carrier extends over at least a portion of the first lateral portion of the frame, and supports and guides portions of the screen extending over the first lateral portion of the frame during the slide mode operation of the screen. The second section of the screen carrier extends over at least a portion of the second panel, and supports and guides portions of the screen extending over the second panel during the slide mode operation of the screen. In embodiments, the first section of the screen carrier extends over at least a portion of the second lateral portion of the frame, and supports and guides portions of the screen extending over the second lateral portion of the frame during the slide mode operation of the screen; and the screen carrier further includes a disengagement structure that causes the screen to disengage from the first section of the screen carrier when the screen is at its pivot ready position.

(45) In embodiments, the first section of the screen carrier is mounted to the sill of the frame and supports and guides a bottom portion of the screen during the slide mode operation of the screen; the second section of the screen carrier is mounted to a top portion of the second panel and supports and guides a top portion of the screen during the slide mode operation of the screen; and the disengagement structure includes one or more ramps that cause the screen to disengage from the first section of the screen carrier when the screen is at its pivot ready position. In embodiments, one or more of the one or more ramps is located on the second section of the screen carrier. In embodiments, the screen includes one or more riders coupling the screen to the second section of the screen carrier; and each of the one or more ramps is configured to engage an associated one of the one or more riders.

(46) In any or all embodiments of the fenestration system of the sixth example, the screen carrier may further include a third section mounted to the head of the frame, to support and guide at least portions of the screen extending over the first lateral portion of the frame during the slide mode operation. The second section and the third section of the screen carrier may be positioned with respect to one another to facilitate the top portion of the screen transitioning between the second section and the third section of the screen carrier during the slide mode operation of the screen.

(47) A seventh example is a fenestration system with a pivot wheel assembly and a head slide. Embodiments of the fenestration system in accordance with the seventh example include a frame defining a frame plane and including a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb; a first panel supported by the frame; a locking pivot rider assembly coupling one of a top or bottom portion of the first panel to the frame and to facilitate slide mode operation and pivot mode operation of the first panel; a slide assembly; and a latch. During the slide mode operation the first panel is in a slide position generally parallel to the frame plane and can slide within the frame between the first and second jambs. During the pivot mode operation the first panel can pivot outwardly to a pivot open position beyond the frame plane. The locking pivot rider assembly may include a rider coupling the first panel to the frame for the slide mode operation; a pivot rotatably coupling the first panel to the rider for the pivot mode operation; and a lock coupled to the rider and the pivot, to releasably lock the rider with respect to the frame. Rotation of the pivot during the pivot mode operation of the first panel from the slide position to the pivot open position actuates the lock and causes the lock to inhibit sliding movement of the rider on the frame. Rotation of the pivot during the pivot mode operation of the first panel from the pivot open position to the slide position releases the lock to allow sliding movement of the rider on the frame. The slide assembly couples the other of the top or bottom of the first panel to the frame and to facilitate the slide mode operation and the pivot mode operation of the first panel. The slide assembly may include a slide body configured for sliding motion between the first and second jambs during the slide mode operation; a pivot rotatably coupling a second end portion of the first panel to a second end portion of the slide body; and a latch releasably coupling a first end portion

of the first panel to a first end portion of the slide body, and operable between a latched position and an unlatched position. When the latch is in the latched position the latch couples the first panel to the slide body to facilitate the slide mode operation of the first panel. When the latch is in the unlatched position the latch releases the first panel from the slide body to facilitate the pivot mode operation of the first panel. The latch actuator is coupled to the latch and operable by a user to cause the latch to move between the latched position and the unlatched position.

(48) Other embodiments of the fenestration system of the seventh example may include any or all features of the first through sixth examples.

(49) An eighth example is a fenestration system with a pivot wheel assembly and a screen carrier. Embodiments of the fenestration system in accordance with the eighth example include a frame defining a frame plane and including a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb; a second panel supported by the frame and configured for pivot mode operation; a first panel supported by the frame and configured for slide mode operation and pivot mode operation with respect to the frame; a locking pivot rider assembly coupling the first panel to the frame and to facilitate the slide mode operation and the pivot mode operation of the first panel; a screen supported by the frame; and a screen carrier coupling the screen to the frame. During the pivot mode operation the second panel can pivot between a pivot closed position generally parallel to the frame plane and a pivot open position beyond the frame plane, wherein the second panel extends over a second lateral portion of the frame when in its pivot closed position. During the slide mode operation the first panel is in a slide position generally parallel to the frame plane and can slide within the frame between slide closed and slide open positions, and during the pivot mode operation the first panel can pivot outwardly beyond the frame plane, and wherein when the first panel is in the slide closed position the first panel extends over a first lateral portion of the frame. The locking pivot rider assembly couples the first panel to the frame to facilitate the slide mode operation and the pivot mode operation of the first panel. The locking pivot rider assembly may include a rider coupling the first panel to the frame for the slide mode operation; a pivot rotatably coupling the first panel to the rider for the pivot mode operation; and a lock coupled to the rider and the pivot, to releasably lock the rider with respect to the frame. Rotation of the pivot during the pivot mode operation of the first panel from the slide position to the pivot open position actuates the lock and causes the lock to engage the frame. Rotation of the pivot during the pivot mode operation of the first panel from the pivot open position to the slide position releases the lock and allows sliding movement of the rider on the frame. The screen carrier couples the screen to the frame and facilitates slide mode operation and pivot mode operation of the screen. During the slide mode operation the screen can slide within the frame between a slide closed position corresponding to the slide closed position of the first panel, a slide open position at least partially overlapping the second panel when the second panel is in its pivot closed position, and a pivot ready position engaged with the second panel when the second panel is in its pivot closed position. During its pivot mode operation the screen is in its pivot ready position and can move outwardly beyond the frame plane with the second panel during the pivot mode operation of the second panel.

(50) Other embodiments of the fenestration system of the eighth example may include any or all features of the first through sixth examples.

(51) A ninth example is a fenestration system with a head slide and screen carrier. Embodiments of the fenestration system of the ninth example include a frame defining a frame plane and including a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb; a second panel supported by the frame and configured for pivot mode operation, wherein during the pivot mode operation the second panel can pivot between a pivot closed position generally parallel to the frame plane and a pivot open position beyond the frame plane, wherein the second panel extends over a second lateral portion of the frame when in its pivot closed position; a first panel supported by the frame and configured for slide mode operation and pivot mode operation with respect to the frame, wherein during the slide mode operation the first panel is in a slide position generally parallel to the

frame plane and can slide within the frame between slide closed and slide open positions, and during the pivot mode operation the first panel can pivot outwardly beyond the frame plane, and wherein when the first panel is in the slide closed position the first panel extends over a first lateral portion of the frame; a slide assembly coupling the first panel to the frame and to facilitate the slide mode operation and the pivot mode operation of the first panel; a latch actuator; a screen supported by the frame; and a screen carrier coupling the screen to the frame. The slide assembly may include a slide body configured for sliding motion between the first and second jambs during the slide mode operation; a pivot rotatably coupling a second end portion of the first panel to a second end portion of the slide body; and a latch releasably coupling a first end portion of the first panel to a first end portion of the slide body, and operable between a latched position and an unlatched position. When the latch is in the latched position the latch couples the first panel to the slide body to facilitate the slide mode operation of the first panel. When the latch is in the unlatched position the latch releases the first panel from the slide body to facilitate the pivot mode operation of the first panel. The screen carrier facilitates slide mode operation and pivot mode operation of the screen. During the slide mode operation the screen can slide within the frame between a slide closed position corresponding to the slide closed position of the first panel, a slide open position at least partially overlapping the second panel when the second panel is in its pivot closed position, and a pivot ready position engaged with the second panel when the second panel is in its pivot closed position. During its pivot mode operation the screen is in its pivot ready position and can move outwardly beyond the frame plane with the second panel during the pivot mode operation of the second panel. The latch actuator may be coupled to the latch and operable by a user to cause the latch to move between the latched position and the unlatched position.

(52) Other embodiments of the fenestration system of the ninth example may include any or all features of the first through sixth examples.

(53) A tenth example is a method for operating a fenestration system including a frame and first and second panels mounted with respect to the frame. Embodiments of the method of the tenth example may comprise sliding the first panel between slide closed and slide fully open positions with respect to the second panel when the second panel is at a closed position; pivoting the second panel from its closed position to a pivot open position when the first panel is at or between its slide fully open and slide closed positions; sliding the first panel to a pivot ready position beyond the slide fully open position and opposite the slide fully open position from the slide closed position after the second panel is pivoted to its pivot open position; pivoting the first panel from its pivot ready position to a pivot open position while simultaneously locking the first panel with respect to the frame to inhibit sliding motion of the first panel; pivoting the first panel from its pivot open position to its pivot ready position while simultaneously unlocking the first panel with respect to the frame to allow sliding motion of the first panel; sliding the first panel from its pivot ready position to a position at or between its slide fully open and slide closed positions; and pivoting the second panel from its pivot open position to its closed position when the first panel is at the position at or between its slide fully open and slide closed positions.

(54) Embodiments of the method of the tenth example further comprise attempting, unsuccessfully because of restraint on sliding motion, to slide the first panel beyond its slide fully open position to its pivot ready position when the second panel is at its closed position.

(55) In any and all embodiments of the tenth example, sliding the first panel to its pivot ready position may include sliding the first panel to its pivot ready position while the first panel is latched for sliding motion with respect to the frame; and the method may further include unlatching the first panel with respect to the frame to facilitate the pivoting of the first panel from its pivot open position when the first panel is at its pivot ready position. Embodiments may further comprise attempting, unsuccessfully because of restraint on pivoting motion, to pivot the first panel with respect to the frame when the first panel is not at its pivot ready position.

(56) In any and all embodiments of the tenth example, pivoting and sliding the first panel, and

pivoting the second panel, comprise applying force to the first and second panels to cause the pivoting and sliding of the first panel and the pivoting of the second panel.

(57) Any and all embodiments of the tenth example may further comprise sliding a screen between a slide closed position and a slide open position with respect to the frame; sliding the screen from its slide open position to a pivot ready position opposite the slide open position from the slide closed position when the second panel is at its closed position, and engaging the screen with the second panel; pivoting the second panel from its closed position to its pivot open position includes pivoting the screen engaged with the second panel; pivoting the second panel from its pivot open position to its closed position includes pivoting the screen engaged with the second panel; and the method further comprises sliding the screen from its pivot ready position to its closed position when the second panel is at its closed position. In embodiments, sliding the screen between its slide closed and slide open positions includes sliding the screen on a guide track; sliding the screen from its slide open position to its pivot ready position comprises disengaging the screen from the guide track; and sliding the screen from its pivot ready position to its closed position comprises engaging the screen with the guide track.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a perspective illustration of a first side of a fenestration unit in accordance with embodiments, showing a first panel, a second panel, a screen, and a frame.
- (2) FIG. 1B is a perspective illustration of a second side of the fenestration unit shown in FIG. 1A, in accordance with embodiments.
- (3) FIG. 1C is a perspective illustration of the second side of the fenestration unit shown in FIG. 1B, without the screen, in accordance with embodiments.
- (4) FIGS. 2A-2B are perspective illustrations of the first and second sides, respectively, of the fenestration unit shown in FIGS. 1A and 1B, but with the first panel, the second panel, and the screen, in various positions with respect to the frame.
- (5) FIGS. 3A-3B are perspective illustrations of the first and second sides, respectively, of the fenestration unit shown in FIGS. 1A and 1B, but with the first panel, the second panel, and the screen, in various positions with respect to the frame.
- (6) FIGS. 4A-4B are perspective illustrations of the first and second sides, respectively, of the fenestration unit shown in FIGS. 1A and 1B, but with the first panel, the second panel, and the screen, in various positions with respect to the frame.
- (7) FIGS. 5A-5B are perspective illustrations of the first and second sides, respectively, of the fenestration unit shown in FIGS. 1A and 1B, but with the first panel, the second panel, and the screen, in various positions with respect to the frame.
- (8) FIGS. 6A-6B are perspective illustrations of the first and second sides, respectively, of the fenestration unit shown in FIGS. 1A and 1B, but with the first panel, the second panel, and the screen, in various positions with respect to the frame.
- (9) FIGS. 7A-7D are detailed perspective illustrations of a bottom portion of the first panel and portions of a sill of the frame, in accordance with embodiments.
- (10) FIGS. 8A-8E are detailed perspective illustrations of a second rider for coupling a bottom portion of the first panel to the sill of the frame, in accordance with embodiments.
- (11) FIG. 9 is a detailed perspective illustration of a bottom portion of the first panel and the sill of the frame, with portions of the first panel removed, illustrating a first rider for coupling the bottom portion of the first panel to the sill, in accordance with embodiments.
- (12) FIGS. 10A and 10B are detailed perspective illustrations of a portion of the sill and a guide track, illustrating portions of the sill and guide track at which the first rider will be located when

the first panel is at the pivot ready position, in accordance with embodiments.

(13) FIGS. **11A-11C** are detailed perspective illustrations of a top portion of the first panel, and portions of a head of the frame and a guide track, illustrating a head slide coupling the top portion of the first panel to the head, in accordance with embodiments.

(14) FIGS. **12-14** are perspective views of the top portion of the first panel, illustrating a latch and pivot of the head slide, and a latch actuator on the first panel, in accordance with embodiments.

(15) FIGS. **15A** and **15B** are detailed perspective illustrations of the latch and the latch actuator, in accordance with embodiments.

(16) FIG. **16** is a perspective illustration of the guide track for the first panel on the head of the frame, illustrating portions that cooperate with the latch, in accordance with embodiments.

(17) FIGS. **17A** and **17B** are perspective illustrations of the first and second sides of the frame and screen, including a top portion of the second panel.

(18) FIG. **18** is a perspective illustration of the sill and bottom portion of the screen and second panel, illustrating a first screen carrier section and guide track.

(19) FIGS. **19A-19C** are perspective illustrations of a top portion of the second panel, illustrating a second screen carrier section.

(20) FIG. **20A** is a perspective illustration of the second side of the frame, a top portion of the second panel, and a top portion of the screen.

(21) FIGS. **20B** and **20C** are perspective illustrations of the frame and top portion of the second panel as shown in FIG. **20A**, but with the first jamb and screen removed.

(22) FIGS. **21A** and **21B** are detailed perspective illustrations of the first and second sides of the bottom portion of the screen, showing riders on the bottom portion of the screen.

(23) FIGS. **22A** and **22B** are detailed perspective illustrations of the first and second sides of the top portion of the screen, showing riders on the top portion of the screen.

(24) FIGS. **23A** and **23B** are detailed perspective illustrations of the top and bottom portions, respectively, of the screen and second panel, illustrating the second panel and screen in their pivot ready positions.

(25) FIGS. **24A** and **24B** are detailed perspective illustrations of the first and second sides of a rider on a second end and top portion of the screen, and an associated disengagement structure.

(26) FIG. **25** is a detailed perspective illustration of the second carrier section, showing a disengagement structure.

(27) FIGS. **26A** and **26B** are detailed perspective illustrations of the second rider, illustrating the pivot.

(28) FIG. **27** is a detailed perspective illustration of a bottom portion of the second end portion of the first panel and the sill of the frame, illustrating the first panel at a slide open position and components of a releasable hook, in accordance with embodiments.

(29) FIG. **28** is a detailed perspective illustration of a bottom portion of the second end portion of the first panel and the sill of the frame, with portions of the second stile of the first panel removed to show the second rider, and illustrating the first panel at the pivot ready position and a releasable hook, in accordance with embodiments.

DETAILED DESCRIPTION

(30) Fenestration units according to the inventive examples, including doors and windows having one or more sliding panels and/or one or more pivoting or hinged panels, optionally including a screen, are adapted to provide for versatility in how they may be opened. An exemplary fenestration unit **10** in the form of a door is illustrated in FIGS. **1A-1C**. As shown, the fenestration unit **10** includes a frame **100** and a plurality of panels such as first panel **200**, a second panel **300**, and a third panel such as screen **400**. The fenestration unit **10** is configured to define a boundary between a first space, for example an interior, and a second space, for example an exterior, of a building or other structure, while providing access through an opening **20** in the fenestration unit between the first and second spaces. FIG. **1A** illustrates a first side of the fenestration unit **10**, for

example a side configured to face the interior space, and FIGS. 1B and 1C illustrate a second side of the fenestration unit **10**, for example a side configured to face the exterior space. The screen **400** is not shown in FIG. 1C, to better illustrate other components of the fenestration unit **10**, including the second panel **300**.

(31) As described in greater detail below, in the illustrated embodiments the second panel **300** is a pivoting or hinged panel that is configured to pivot or hinge with respect to the frame **100** between a closed position that is within and generally parallel to the frame, and a range of open positions at which the hinged panel extends beyond or outside of the frame (e.g., to operate in a pivot or hinge mode). The second panel **300** is shown in its closed position in FIGS. 1A-1C. When in its closed position, the second panel **300** extends over, and in the illustrated embodiments covers, a second lateral portion of the frame **100**.

(32) In the illustrated embodiments, the first panel **200** is a sliding panel that is configured to operate with respect to the frame **100** in both a sliding mode and a pivot mode. When in the sliding mode, the first panel **200** is in a slide orientation or position within and generally parallel to the frame **100**, and can slide between a slide closed position, a range of slide open positions including a slide fully open position, and a pivot ready position that is opposite or beyond the slide fully open position from the slide closed position. When the second panel **300** is in its open position and the first panel **200** is in its pivot ready position, the first panel can be operated in the pivot mode to pivot between the pivot ready position and a range of pivot open positions at which the second panel extends beyond or outside of the frame **100**. The first panel **200** is shown in an open position in FIGS. 1A-1C, at which it is parallel to and partially overlaps the second panel **300**. When in its closed position (e.g., shown in FIGS. 6A and 6B), the first panel **200** extends over, and in the illustrated embodiments covers, a first lateral portion of the frame **100** that is at least partially different than the second lateral portion of the frame that the second panel **300** extends over. Although not shown, in embodiments the fenestration unit **10** includes structures, such as for example clips, that cause the first panel **200** and the second panel **300** to interlock when the first and second panels are in their fully closed positions. These structures can, for example, effectively prevent the second panel **300** from moving toward its open position from its closed position when the first panel **200** is at its fully closed position.

(33) In embodiments, the screen **400** is a sliding screen that is configured to operate with respect to the frame **100** and the second panel **300** in both a sliding mode and a pivot mode. When in the sliding mode, the screen **400** is in a slide orientation or position within and generally parallel to the frame **100**, and can slide between a slide closed position, a range of slide open positions, and a hinge- or pivot ready position. When the screen **400** is in its slide closed position, it may extend across the first lateral portion of the frame **100** corresponding to the location of the first panel **200** when the first panel is in its slide closed position (e.g., shown in FIGS. 5A and 5B). The first panel **200** may also be opened when the screen **400** is in its slide closed position, with the screen thereby providing a screened opening through the frame **100**. When the screen **400** is in any of its slide open positions (with the first panel **200** in any of its slide closed or slide open positions), the screen at least partially overlaps the second panel **300** when the second panel is in its closed position (e.g., extends over at least a portion of the second lateral portion). When the screen **400** is in its pivot ready position, it is engaged with or coupled to the second panel **300**, and disengaged from the frame **100**, thereby allowing the screen to move with the second panel (e.g., to swing outwardly) when the second panel pivots or hinges between its open and closed positions.

(34) The illustrated embodiments of the frame **100** include a sill **110**, a head **112**, a first jamb **114** and a second jamb **116** that generally define a frame opening **115**. The sill **110** is a structural frame element that generally defines a bottom portion of the frame **100** and fenestration unit **10**. The head **112** is a structural frame element that generally defines a top portion of the frame **100** and the fenestration unit **10**. The sill **110** and head **112** each generally extend laterally between the first and second jambs **114** and **116**. The first and second jambs **114** and **116** are structural frame elements

that extend vertically between the sill **110** and head **112**, and generally define first and second end portions of the frame. In embodiments, the head **112** may be supported by each of the first and second jambs **114** and **116**.

(35) The illustrated embodiments of the first panel **200** include a bottom rail **210**, a top rail **212**, a first stile **214** and a second stile **216**. The bottom rail **210**, top rail **212**, first stile **214** and second stile **216** collectively define a frame **218** of the first panel **200**. In embodiments, an insert **220**, such as a glass panel insert, is mounted within the frame **218**. It will be appreciated that inserts **220** other than glass inserts, such as for example wood, plastic and/or metal, may be mounted to the frame **218**. It will also be appreciated that the insert **220** need not be a structure that is impervious to atmospheric substances such as air, and may for example be a screen that permits air flow therethrough. Yet other embodiments of the first panel **200** (not shown) do not include a separate frame such as **218** (e.g., are frameless). The bottom rail **210**, top rail **212**, first stile **214** and second stile **216** generally define a bottom portion, a top portion, a first end portion and a second end portion, respectively, of the first panel **200**.

(36) The illustrated embodiments of the second panel **300** include a bottom rail **310**, a top rail **312**, a first stile **314** and a second stile **316**. The bottom rail **310**, top rail **312**, first stile **314** and second stile **316** collectively define a frame **318** of the second panel **300**. In embodiments, an insert **320**, such as a glass panel insert, is mounted within the frame **318**. It will be appreciated that inserts **320** other than glass inserts, such as for example wood, plastic and/or metal, may be mounted to the frame **318**. It will also be appreciated that the insert **320** need not be a structure that is impervious to atmospheric substances such as air, and may for example be a screen that permits air flow therethrough. Yet other embodiments of the second panel **300** (not shown) do not include a separate frame such as **318** (e.g., are frameless). The bottom rail **310**, top rail **312**, first stile **314** and second stile **316** generally define a bottom portion, a top portion, a first end portion and a second end portion, respectively, of the second panel **300**.

(37) The second panel **300** is mounted to the frame **100** by pivoting structures such as hinges **330** to enable the pivoting or hinging opening and closing of the second panel. The illustrated embodiment includes three hinges **330** coupling the second stile **316** of the second panel **300** to the second jamb **116** of the frame **100**. Other embodiments (not shown) include other structures for mounting the second panel **300** for hinging or pivoting operation with respect to the frame **100**.

(38) Embodiments of the screen **400** may include a bottom rail **410**, a top rail **412**, a first stile **414** and a second stile **416**. In embodiments, the bottom rail **410**, top rail **412**, first stile **414** and second stile **416** collectively define a frame **418** of the screen **400**. In embodiments, an insert **420**, such as a screen panel insert, is mounted within the frame **418**. Although described as a screen **400** in the illustrated embodiments, it will be appreciated the third panel may take other forms, such as for example those of the first panel **200** and the second panel **300** described above. The bottom rail **410**, top rail **412**, first stile **414** and second stile **416** generally define a bottom portion, a top portion, a first end portion and a second end portion, respectively, of the screen **400**.

(39) FIGS. 1A-1C show the first panel **200** configured for slide mode operation and in a slide open position, the second panel **300** in its closed position, and the screen **400** (FIG. 1B) in its pivot ready position engaged with the second panel **300** and disengaged from the frame **100**. With the first panel **200**, second panel **300** and screen **400** in these positions, an opening **20** is defined in the frame **100** (between the first stile **214** of the first panel **200** and the first jamb **114** of the frame), permitting passage through the fenestration unit **10**.

(40) FIG. 2A illustrates the first side of the fenestration unit **10** with the second panel **300**, and the screen **400** coupled to the second panel, hinged or pivoted outwardly to their open positions outside of the frame **100**. In FIG. 2A the first panel **200** is at the same open position shown in FIGS. 1A-1C. FIG. 2B illustrates the second side of the fenestration unit **10** with the first panel **200**, second panel **300** and screen **400** at the positions shown and described above in connection with FIG. 2A.

(41) FIG. 3A illustrates the first side of the fenestration unit **10** with the second panel **300** and

screen **400** at their hinged or pivot open positions shown and described above in connection with FIGS. **2A** and **2B**, and the first panel **200** in its slide mode and at its pivot ready position. FIG. **3B** illustrates the second side of the fenestration unit **10** with the first panel **200**, second panel **300** and screen **400** at the positions shown and described above in connection with FIG. **3A**.

(42) FIG. **4A** illustrates the first side of the fenestration unit **10** with the second panel **300** and screen **400** at their hinged or pivot open positions shown and described above in FIGS. **2A** and **2B**, and the first panel **200** at a pivot open position following operation in the pivot mode. FIG. **4B** illustrates the second side of the fenestration unit **10** with the first panel **200**, second panel **300** and screen **400** at the positions shown and described above in connection with FIG. **4A**. With the first panel **200**, second panel **300** and screen **400** in these positions, an opening **20'** is defined in the frame **100** (between the first panel and the jamb **114** of the frame). The opening **20'** is larger than the opening **20** (e.g., as shown in FIG. **1A**) because both the first panel **200** and second panel **300** (with the screen **400** attached thereto) are swung open outside of the frame **100**.

(43) FIG. **5A** illustrates the first side of the fenestration unit **10** with the first panel **200** configured for operation in its slide mode and at a slide open position, and the second panel **300** in its closed position (e.g., as in FIGS. **1A** and **1B**), but with the screen **400** configured for slide mode operation and at its slide closed position. The screen **400** thereby provides a screen barrier across the opening **20** between the first panel **200** and the frame **100**. FIG. **5B** illustrates the second side of the fenestration unit **10** with the first panel **200**, second panel **300** and screen **400** at the positions shown and described above in connection with FIG. **5A**.

(44) FIG. **6A** illustrates the first side of the fenestration unit **10** with the first panel **200** configured for slide mode operation and at its slide closed position, the second panel **300** in its closed position, and the screen **400** in an open position. FIG. **6B** illustrates the second side of the fenestration unit **10** with the first panel **200**, second panel **300** and screen **400** at the positions shown and described above in connection with FIG. **6A**. In the illustrated embodiments the entire opening along the length of the frame **100** (e.g., between the first jamb **114** and the second jamb **116**), including the first lateral portion and the second lateral portion, are closed by the first panel **200** and the second panel **300** when the first and second panels are in these positions shown in FIGS. **6A** and **6B**. Although not shown, the screen **400** can be operated in its slide mode and positioned at a range of slide open positions between its slide closed position shown in FIGS. **5A** and **5B**, and its pivot ready position shown in FIG. **1B**.

(45) The illustrated embodiments of fenestration system **10** include one first panel **200** configured to operate in the slide mode and pivot mode, and one second panel **300** configured to pivot between open and closed positions. And as noted above, when the first panel **200** is in its slide closed position and the second panel **300** is in its closed position, the first and second panels extend over the first and second lateral portions of the frame **100**, and effectively close the opening defined by the frame. In the illustrated embodiments the first panel **200** and the second panel **300** are parallel and transversely offset (e.g., about an axis perpendicular to the major planar surfaces of the panels) with respect to one another. The first panel **200** and the second panel **300** also overlap one another (e.g., adjacent the second stile **216** of the first panel and the first stile **314** of the second panel) when the first panel is in its slide closed position and the second panel is in its closed position as shown in the illustrated embodiments. Similarly, the screen **400** is parallel to and transversely offset with respect to the first panel **200** and the second panel **300** when the first panel is in its slide closed position and the second panel **300** is in its closed position. However, it will be appreciated that other embodiments may include other configurations panels such as the first and second panels and the screen. Other embodiments may also include more than one first panel, more than one second panel, and/or more than one screen.

(46) FIGS. **7A-7D** are detailed illustrations of a bottom portion of the first panel **200**, including the bottom rail **210** and the first and second stiles **214** and **216**, on a portion of the sill **110** of the frame **100**. As shown, an elongated guide track **40** is located on the side of the sill **110** facing the frame

opening **115**. In embodiments, the guide track **40** may be a member mounted to the sill **110**. In other embodiments the sill **110** may be a molded or extruded member including the guide track **40** integrally formed therein. Although the guide track **40** is shown as including a raised rail **41** extending from the sill **110** in the illustrated embodiments, the guide track **40** may take other forms in other embodiments (not shown). A first rider **500** and a second rider **502** are mounted to the bottom portion of the first panel **200** at spaced-apart locations, and are configured to support the first panel with respect to the sill **110** and guide track **40**, and to cooperate with the sill and guide track to facilitate the slide mode operation and the pivot mode operation of the first panel with respect to the sill and guide track. The first rider **500** and second rider **502** are mounted within the frame **218** of the first panel **200** in the illustrated embodiments, and are shown in phantom in FIGS. **7A-7D**. In the illustrated embodiments, the first rider **500** is mounted to the first panel **200** at a location on the bottom portion and first end portion of the first panel, and is shown for purposes of example in the first stile **214**. Similarly, in the illustrated embodiments the second rider **502** is mounted to the first panel **200** at a location on the bottom portion and second end portion of the first panel, and is shown for purposes of example in the second stile **216**. The first rider **500** and the second rider **502** are guided by the guide track **40** during the slide mode operation of the first panel **200**, and provide smooth and relatively low friction sliding movement of the first panel **200** during the slide mode operation of the first panel.

(47) FIGS. **8A-8E** are detailed perspective illustrations of the second rider **502** and associated portions of the sill **110** and guide track **40** at which the second rider will be located when the first panel **200** (not shown in FIGS. **8A-8D**) is configured for slide mode operation and located at its pivot ready position. The second rider **502** is shown isolated from the first panel **200** in FIGS. **8A-8D**. The first panel **200** is shown with portions removed in FIG. **8E**.

(48) The second rider **502** is a locking pivot rider assembly in the illustrated embodiments, and includes a rider **510**, a pivot **512** a lock **514** and a height adjuster **516**. With the possible exception of a lower portion of the rider **510** that engages the guide track **40**, the illustrated embodiments of the second rider **502** are configured to be mounted within a bore or other recess in the first panel **200**, such as for example in the second stile **216**. The illustrated embodiments of the rider **510** include a body **520** having a bottom portion **522** that is configured to engage and slide on the sill **110** while being guided by the guide track **40**. As described in greater detail below, the body **520** is rotatably mounted to the first panel **200** to facilitate the pivot mode operation of the first panel. The illustrated embodiment of the rider **510** includes a wheel **524** that is rotatably mounted to the body **520** and positioned to engage and roll on the rail **41** of the guide track **40** during the slide mode operation of the first panel **200**. Other embodiments of the rider **510** may include different and/or other features to provide the functions, including the sliding and guiding functions, described herein.

(49) The pivot **512** is rotatably mounted to the rider **510** (e.g., to the body **520** in the illustrated embodiments). The illustrated embodiments of the pivot **512** include a bearing plate **530** mounted to a first or upper end of a rod **532** that extends from a top or upper portion **534** of the body **520**. The bottom portion of the rod **532** is mounted to and extends from the upper portion **534** of the body **520**. The bearing plate **530** may be fixedly mounted to the first panel **200**, for example by screws or other fasteners (not shown) that may extend through bores **536** in the bearing plate and into the first panel.

(50) FIGS. **26A** and **26B** are detailed illustrations of the second rider **502** with portions removed to illustrate the pivot **512**. In the illustrated embodiments the rod **532** extends from the body **520** of the rider **510** and includes a tapered end **533** on its end opposite the rider. The bearing plate **530** engages and rotates on the tapered end **533** of the rod **532** in these embodiments. As described in greater detail below, in the illustrated embodiments the pivot **512** also includes structures that facilitate the functionality of the lock **514** and the height adjuster **516**. In particular, and as shown in FIGS. **26A** and **26B**, at least the lower portion of the rod **532** may be a threaded member that is

threadedly and rotatably mounted in a threaded bore in the body **520** of the rider **510**. A bushing **535** mounted to the rod **532** facilitates mounting of one or more components of the lock **514** to the rod **532**. Because the bearing plate **530** is rotatably mounted to the body **520** (e.g., by the rod **532**), during the pivot mode operation of the first panel **200**, the first panel and the bearing plate can pivot or rotate with respect to the body while the body remains engaged with sill **110** and guide track **40**.

(51) Referring back to FIGS. **8A-8E**, the illustrated embodiments of lock **514** include cam body **540** and one or more lock pins **542** (two are shown in the illustrated embodiments). Each of the lock pins **542** is slidably mounted within a respective bore **544** extending through the body **520**, and between and through the upper portion **534** and the bottom portion **522** of the body. Each of the lock pins **542** is moveable between a retracted position shown in FIGS. **8A-8C** at which the lower end of the lock pin is disengaged from the sill **110**, and an extended position shown in FIG. **8D** at which the lower end of the lock pin is engaged with the sill. Embodiments may include biasing members, such as for example leaf or coil springs (not shown) to bias each of the lock pins **542** toward its retracted position. In the illustrated embodiments, the upper portion **534** of the body **520** through which the lock pins **542** extend is convex in shape.

(52) The cam body **540** is rotatably mounted with respect to the body **520** of the rider **510**, above the upper portion **534** of the rider body. In the illustrated embodiments, for example as shown in FIG. **8B**, a hex nut **533** is threaded onto the rod **532** (and can therefore rotate with respect to the rod **532**, and seats within a hex-shaped recess in the upper portion of the cam body **540**. The cam body **540** is coupled to the pivot **512** (e.g., by the height adjuster **516**, adjustment mechanism **517**, including key **556**, and rod **532** as described in greater detail below), and rotates with respect to the body **520** of rider **510** during the rotation of the first panel **200** during its pivot mode operation.

(53) A lower portion of the cam body **540** includes a cam surface **550** that includes retract portions **552** and extend portions **554** configured to engage the lock pins **542** as the cam body rotates. As noted above, the cam body **540** is configured to rotate with the bearing plate **530** of the pivot **512** and the first panel **200** when the first panel **200** is operated in its pivot mode.

(54) The body **520** of the rider **510** and cam body **540** are mounted with respect to one another so that when the first panel **200** is configured to operate in its slide mode, the cam body is positioned with the retract portions **552** of the cam surface **550** adjacent to the lock pins **542** as shown in FIGS. **8A-8C**. The lock pins **542** can then be in their retracted positions, disengaged from the sill **110** and allowing the second rider **502** and the first panel **200** to slide between the slide closed and pivot ready positions during the slide mode operation.

(55) When the first panel **200** is rotated from the pivot ready position to the pivot open position, the rotation will cause the cam body **540** to rotate with respect to the body **520** of the rider **510**, with the cam surface **550** engaging the lock pins **542** and forcing the lock pins to their extended positions. The lock pins **542** will be in their fully extended positions when the cam body **540** is positioned with the extend portions **554** of the cam surface **550** adjacent to the lock pins as shown in FIG. **8D**.

(56) In embodiments, the sill **110** may include recesses such as **568** that are configured and located to receive the lock pins **542** in the extended positions when the first panel **200** is at the pivot ready position. Although the lock pins **542** releasably engage the sill **110** in the illustrated embodiments, other embodiments (not shown) of the lock include additional or alternative structures to lock movement of the second rider **502** with respect to the sill. For example, in embodiments with a rotating wheel, the lock may releasably lock the wheel with respect to the structure to which it is rotatably mounted, to prevent rotation of the wheel.

(57) FIGS. **8E**, **27** and **28** illustrate a releasable hook **580** that may be incorporated into embodiments of the fenestration system **10** to facilitate the pivot mode operation of the first panel **200**. Because of the configuration of the cam surface **550** (e.g., the sloping surface between the retract portions **552** and the extend portions **554**), during initial portions of the rotation of the first

panel **200** (and the cam body **540**) when the first panel is rotated from its pivot ready position toward the pivot open position during the pivot mode operation, the lock pins **542** are moved toward the sill **110**, but may not have been moved a sufficient distance toward their extended positions to engage the sill. The lock **514** therefore may not effectively lock the second rider **502** to the sill **110** during these initial portions of the pivot mode operation. Similarly, the lock **514** may effectively release the second rider **502** from the sill **110** during final portions of the rotation of the first panel **200** back to its pivot ready position during the pivot mode operation. To prevent sliding movement of the first panel **200** along the sill **110** during these initial and final portions of rotation during the pivot mode operation, and to position the first panel **200** at the pivot ready position, the illustrated embodiments of fenestration unit **10** include the releasable hook **580**.

(58) In the illustrated embodiments, the releasable hook **580** includes a panel member **582** and a sill member **584**. The panel member **582** includes a stop surface **583** adjacent the end edge portion of the second stile **216** of the first panel **200**, an extending portion **586** that extends from the end edge portion of the second stile **216**, and a lip **588** that extends toward the sill **110** from the extending portion at a location spaced from the second stile. The sill member **584** includes a lip **590** that extends upwardly, away from the sill **110**. As perhaps best shown in FIGS. **27** and **28**, the lip **588** of the panel member **582** extends only partially across a thickness of the end edge of the second stile **216**, and is located toward the second side of the first panel **200**. The lip **590** of the sill member **584** is complementary to the lip **588** of the panel member **582**, and extends only partially across a width of the sill **110**, and is located on the side of the sill toward the first side of the first panel **200**.

(59) As the first panel **200** is moved from a slide open position such as that shown in FIG. **27** to the pivot ready position shown in FIG. **28**, the stop surface **583** of the panel member **582** engages a stop surface **585** on a side of the lip **590** facing the first panel **200**. The stop surfaces **583** and **585**, which in embodiments are complimentary curved surfaces, are located and configured so that the first panel **200** is at the pivot ready position when the surfaces are engaged. During this movement of the first panel **200**, the lip **588** of the panel member **582** passes by the lip **590** of the sill member **584**. During the initial and final portions of the rotation of the first panel **200** during the pivot mode operation, the lip **588** of the panel member **582** moves past the lip **590** of the sill member **584** on the side of the sill member opposite the first panel. As shown, the lips **588** and **590** may be curved to facilitate the relative movement of these members during the pivot mode operation of the first panel **200**. Although facilitating rotation of the first panel **200** with respect to the sill **110** during these initial and final portions of rotation of the first panel during the pivot mode operation, the panel member **582** and sill member **584** of the releasable hook **580** cooperate to prevent the first panel from sliding on the sill when the lock pins **542** of the lock **514** are not engaged with the sill **110**.

(60) Referring back to FIGS. **8A-8E**, **26A** and **26B**, the illustrated embodiments of height adjuster **516** include an adjustable mount of the pivot **512** to the rider **510**, and an adjustment mechanism **517** for actuating the adjustable mount. The adjustable mount facilitates the height adjustment of the pivot **512** with respect to the rider **510**, and in the illustrated embodiments is provided by the threaded coupling of the rod **532** to the body **520** of the rider **510**. Rotation of the rod **532** causes the rod to extend further out of the body **520**, or to retract further within the body, thereby adjusting the height or distance of the bearing plate **530** with respect to the rider **510**. In the illustrated embodiments, the adjustment mechanism **517** is operable to rotate the rod **532**, and thereby cause the height adjustment.

(61) The illustrated embodiments of the adjustment mechanism **517** includes a body **571**, an annular gear **570**, a key **556**, and a pinion gear **572**. Body **571** is rotatably mounted to the rod **532** via a bushing **535**. The bushing **535** is fixedly mounted to the rod **532**, and supports the body **571**. The annular gear **570** is fixedly mounted to the rod **532**, and is therefore fixedly mounted with respect to the bushing **535**. Annular gear **570** and rod **532** can thereby rotate within the body **571** and the cam body **540** during actuation of the adjustment mechanism **517**. Key **556** is a generally

planar member, but curved to match the radii of the components to which it is coupled in the illustrated embodiments, and is fixedly mounted to the body **571**. In the illustrated embodiments the key **556** is engaged in a cutout or slot of a flange or shoulder **558** that extends from a central portion of the body **571**. The pinion gear **572** is rotatably mounted to both the key **556** and the central portion of the body **571** about an axis that extends generally perpendicular to the longitudinal axis of the rod **532**. The pinion gear **572** engages the annular gear **570**. A portion of the key **556** is slidably engaged with the cam body **540** of the lock **514** to facilitate motion of the key upwardly and downwardly with respect to the cam body in directions generally parallel to the rod **532**, while being retained at a fixed position about the circumference of the cam body. In the illustrated embodiments, this functionality is provided by locating portions of the key **556** in a cutout or slot of a flange or shoulder **560** that extends from the cam body **540**.

(62) Annular gear **570** and pinion gear **572** comprise a bevel gear set in the illustrated embodiments. Rotation of the pinion gear **572**, for example by a screwdriver engaged with the Phillips-head slot in the head **574** of the pinion gear, causes rotation of the rod **532** of the pivot **512** via the rotation of the annular gear **570**. This rotation of the pivot **512** causes the pivot and its bearing plate **530** to extend or retract with respect to the body **520** of rider **510** and the cam body **540**, and thereby raises or lowers the first panel **200** with respect to the rider and the sill **110**. Key **556** is configured to prevent rotation of the body **571** of the adjustment mechanism **517** with respect to the cam body **540** of the lock **514** during actuation of the adjustment mechanism. However, because the key **556** can slide with respect to the cam body **540**, the key facilitates movement of the body **571**, annular gear and pinion gear as the pivot **512** extends and retracts with respect to the rider **510** during actuation of the adjustment mechanism **517**. In the illustrated embodiments, the bearing plate **530** is fixedly mounted to the first panel **200**, and the actuation of the adjustment mechanism **517** causes the rider **510** to move with respect to the first panel **200**.

(63) During the rotation of the first panel **200** during its pivot mode operation, the rod **532** and the cam body **540** mounted thereto rotate with respect to the body **520** of the rider **510** by the treaded coupling of the body **520** to the rod **532**. Because of this relative movement between the cam body **540** and the body **520** of the rider **510** along the axis of the rod **532** during the pivot mode operation of the panel **200**, the cam body and the body of the rider are configured with sufficient space between them to accommodate the relative movement during the pivot mode operation of the first panel.

(64) As shown for example in FIGS. 7D and 8E, the head **574** of the pinion gear **572** may be accessed through an opening **573** through the end edge of the second stile **216** so that the adjustment mechanism **517** of the height adjuster **516** can be actuated from outside of the first panel **200**. The height adjuster **516** may be actuated to adjust the height of the second end portion of the first panel **200** during set up and adjustment operations of the first panel. Other embodiments of the second rider **502** (not shown) include other height adjustment mechanisms or structures, or no height adjuster.

(65) FIG. 9 is a perspective illustration of the bottom and first end portions of the first panel **200** and adjacent portions of the sill **110** and guide track **40**, with portions of the first stile **214** removed to illustrate the first rider **500**. In the illustrated embodiments, portions of the first rider **500** are substantially the same as or similar to the second rider **502** described above in connection with FIGS. 8A-8E, and similar features are identified by similar reference numbers. As shown, first rider **500** is an assembly that includes a rider **510'** and a height adjuster **516'**. With the possible exception of a lower portion of the rider **510'** that engages the guide track **40**, the illustrated embodiment of the first rider **500** is configured to be mounted with a bore or other recess in the first panel **200**, such as for example in the first stile **214**. The illustrated embodiments of the first rider **500** includes a body **520'** having a bottom or lower portion **522'** that is configured to engage and slide on the sill **110** while being guided by the guide track **40**. The illustrated embodiment of the body **520'** includes a wheel **524'** that is rotatably mounted to the body and positioned to engage and roll on the

rail **41** of the guide track **40** during the slide mode operation of the first panel **200**.

(66) The height adjuster **516'** of the first rider **500** can be substantially the same as or similar to that of the height adjuster **516** of the second rider **502**, and includes an adjustable mount of the pivot **512'** to the rider **510'**, and an adjustment mechanism **517'** for actuating the adjustable mount. The adjustable mount facilitates the height adjustment of the pivot **512'** with respect to the rider **510'**, and in the illustrated embodiments is provided by the threaded coupling of the rod **532'** to the body **520'** of the rider **510'**. Rotation of the rod **532'** causes the rod to extend further out of the body **520'**, or to retract further within the body, thereby adjusting the height or distance of the bearing plate **530'** with respect to the rider **510'**. In the illustrated embodiments, the adjustment mechanism **517'** is operable to rotate the rod **532'**, and thereby cause the height adjustment.

(67) The illustrated embodiments of the adjustment mechanism **517'** include a body **571'**, an annular gear **570'**, a key **556'**, and a pinion gear **572'**. Body **571'** is rotatably mounted to a bushing (not shown in FIG. 9). The bushing is fixedly mounted to the rod **532'**. The annular gear **570'** is fixedly mounted to the rod **532'** and the bushing. Annular gear **570'** and rod **532'** can thereby rotate within the body **571'** during actuation of the adjustment mechanism **517'**. Key **556'** is fixedly mounted to the body **571'**. In the illustrated embodiments the key **556'** is engaged in a cutout or slot of a flange or shoulder **558'** that extends from a central portion of the body **571'**. The pinion gear **572** is rotatably mounted to both the key **556'** and the central portion of the body **571'** about an axis that extends generally perpendicular to the longitudinal axis of the rod **532'**. The pinion gear **572'** engages the annular gear **570'**. A portion of the key **556'** is fixedly coupled to the first panel **200**, for example by a screw, bolt or other fastener (not shown) that attaches the key to the first stile **314**.

(68) Rotation of the pinion gear **572'**, for example by a screwdriver engaged with the Phillips-head slot in the head **574'** of the pinion gear, causes rotation of the rod **532'** of the pivot **512'** via the rotation of the annular gear **570'**. This rotation of the pivot **512'** causes the body **520'** of the rider **510'** to extend or retract with respect to the pivot and its bearing plate **530'**, and thereby raises or lowers the first end portion of the first panel **200** with respect to the rider and the sill **110**. Key **556'** is configured to prevent rotation of the body **571'** and pinion gear **572'** of the adjustment mechanism **517'** with respect to the rider **510'** during actuation of the adjustment mechanism, but because the key can move with respect to the rider **510'**, the key facilitates movement of the body **571'**, annular gear and pinion gear as the pivot **512'** extends and retracts with respect to the rider **510'**.

(69) As shown for example in FIGS. 7D and 8E, the head **574'** of the pinion gear **572'** may be accessed through an opening **573'** through the end edge of the first stile **214** so that the adjustment mechanism **517'** of the height adjuster **516'** can be actuated from outside of the first panel **200**. The height adjuster **516'** may be actuated to adjust the height of the first end portion of the first panel **200** during set up and adjustment operations of the first panel. Other embodiments of the first rider **502'** (not shown) include other height adjustment mechanisms or structures, or no height adjuster. Although the first rider **502'** couples the bottom portion of the first panel **200** to the sill **110** in the illustrated embodiments, in other embodiments the first rider may couple the top portion of a panel such as **200** to the frame **100**.

(70) FIGS. 10A and 10B illustrate portions of the sill **110** and guide track **40** at which the first rider **500** of the first panel **200** will be located when the first panel is at the pivot ready position. When the first panel **200** is at the pivot ready position and the first panel is pivoted outwardly during the pivot mode operation, portions of the first panel, including the first end portion and the first stile **214**, swing outwardly beyond the frame **100**. To facilitate this pivot mode operation, the guide track **40**, and in the illustrated embodiments the rail **41**, include a gap **42** located at the position where the first rider **500** will be positioned when the first panel **200** is at its pivot ready position.

(71) In the illustrated embodiment the gap **42** in the guide track **40** is defined by a wheel stop **44** on the sill **110**. In the illustrated embodiments the wheel stop **44** is a raised projection or wall that is transversely offset from the gap **42** in a direction opposite the gap from the side of the sill **110** that

the first rider **500** moves toward during the pivot mode operation (e.g., toward the first side of the fenestration unit **10** from the guide track **40**). The wheel stop **44** is positioned to engage the rider **500** when the rider is aligned with the guide track (e.g., when the wheel **524'** of the rider is aligned with the rail **41**). The wheel stop **44** therefore prevents the first panel **200** from being moved beyond the position at which it is aligned with the guide track **40** and configured for slide mode operation when the first panel is being closed during pivot mode operation. Projections **43** on the opposite ends of the gap **42**, and at positions transversely offset from the gap in a direction opposite the gap from the wheel stop **44** (e.g., toward the second side of the fenestration unit **10** from the guide track **40**), engage the first rider **500** and inhibit slide mode operation of the first panel **200** if the first panel is not fully moved to the position at which it is aligned with the guide track **40** and configured for slide mode operation when the first panel is being closed during pivot mode operation. Because of the gap **42** and its location on the guide track **40**, when the first panel **200** is at its pivot ready position, the first rider **500**, including its wheel **524'**, are disengaged from the guide track **40**, thereby enabling the first panel to operate in its pivot mode. In embodiments, when the first panel **200** is at other positions along the frame **100** during its slide mode operation, including positions between its slide closed and slide fully open positions, the first rider **500** is engaged with the guide track **40**, thereby inhibiting pivot mode operation of the first panel.

(72) FIG. 1A illustrates stop members **45** (two are shown for purposes of example) that are configured to engage the first panel **200** when the first panel reaches the slide fully open position during slide mode operation, and to prevent the first panel from sliding beyond the slide fully open position when the second panel **300** is in its closed position. Stop members **45** are mounted to the second panel **200** in the illustrated embodiments, and engage the second end edge of the first panel **200** when the first panel is at the slide fully closed position. Other embodiments of the fenestration unit **10** include other configurations and/or structures that provide functions of the stop members **45**.

(73) In the illustrated embodiments, the stop members **45** function as travel limits, and inhibit motion of the first panel **200** from the slide fully closed position toward the pivot ready position when the second panel **300** is at the closed position. The first panel **200** is thereby effectively prevented from being moved to the pivot ready position, and effectively prevented from being operated in its pivot mode, while the second panel **300** is closed. However, the stop members **45** are configured so that when the second panel **300** is rotated to a pivot open position, the stop members no longer engage the first panel **200** during its slide mode operation, and allow or facilitate sliding motion of the first panel beyond the slide fully open position to the pivot ready position. The first panel **200** can then be operated in its pivot mode when the second panel **300** is in a pivot open position. In embodiments, the stop members **45** are elongated members mounted at an end thereof to the second panel **300**. The stop members **45** may be flexible or otherwise configured to resist breakage, for example when engaged by the first panel **200** during the operation of the fenestration unit **10** (e.g., if the second panel **300** is closed when the first panel **200** is at its pivot ready position).

(74) FIGS. 11A-11C are detailed illustrations of a top portion of the first panel **200**, including the top rail **212** and the first and second stiles **214** and **216**, on the head **112** of the frame **100**. In particular, FIG. 11A shows the first side of the portions of the first panel **200** and head **112**. FIG. 11B shows the portions of the first panel **200** and head **112** from the end of the first panel looking toward the first stile **214**. FIG. 11C shows the portions of the first panel **200** and head **112** from the end of the first panel looking toward the second stile **216**. As shown, an elongated channel-type guide track **50** is located on the side of the head **112** facing the frame opening **115**. In embodiments, the guide track **50** may be a member mounted to the head **112**. In other embodiments, the head **112** may be a molded or extruded member including the guide track **50** integrally formed therein.

(75) A head slide **600** couples the top portion of the first panel **200** to the head **112** and guide track

50, and facilitates operation of the first panel in both its slide mode and pivot mode. The head slide **600** is an assembly in the illustrated embodiments and includes a body **602**, a latch **604** that releasably couples the first end portion of the top portion of the first panel **200** (e.g., a portion adjacent to the first stile **214**) to the body, and a pivot **606** that pivotally couples the second end portion of the top portion of the first panel (e.g., a portion adjacent to the second stile **216**) to the body. The body **602** of the head slide **600** is configured to cooperate with the head **112** and guide track **50** to slide within and be guided by the guide track **50**. As shown for example in FIGS. **11B** and **11C**, in the illustrated embodiments the cross-sectional shape and size of the body **602** are complimentary to the cross-sectional shape and size of the guide track **50** to facilitate the sliding and guided motion of the body within the guide track. When the first panel **200** is latched to the head slide **600** by the latch mechanism **604**, the first panel is configured for slide mode operation by the cooperation of the body **602** and the guide track **50**. When the first panel **200** is at its pivot ready position, the latch **604** can be actuated to disengage or release the first panel **200** from the head slide **600**, thereby allowing the first panel to pivot about the pivot **606** and facilitate the pivot mode operation of the first panel. In the illustrated embodiments, the body **602** extends across all or substantially all of an entire width of the first panel **200**, between and including the first stile **214** and the second stile **216**, and includes seals such as **607** (FIG. **12**) to provide an environmental seal between the adjacent edges of the top portion of the first panel and a bottom portion of the body. (76) Head slide **600**, including latch **604** and pivot **606**, and a latch actuator **610**, can be described with reference to FIGS. **12-14**, **15A** and **15B**. FIG. **12** is an illustration of the top portion of the first panel **200**, and the head slide **600**, showing the end edge of the first stile **214**. As shown, in the illustrated embodiments the pivot **606** extends from the body **602** of the head slide **600** into the top portion of the second stile **216** to pivotally couple the head slide to the first panel **200**. The latch actuator **610** is shown at a location adjacent to the end edge of the first stile **214** in the illustrated embodiments, and includes a slide handle **612** that can be actuated by a user to latch and unlatch the latch **604**.

(77) FIG. **13** is an illustration of the top portion of the first panel **200** and the head slide **600** as shown in FIG. **12**, but with end edge portions of the first stile **214** removed to show a latch linkage **614** of the latch actuator **610** that couples the slide handle **612** to the latch **604**. As shown, the latch linkage **614** is configured for reciprocal sliding motion on the first panel **200** in response to actuation of the slide handle **612** (e.g., in upwardly and downwardly directions in the illustrated embodiments). FIG. **14** is an illustration of the top portion of the first panel **200** as shown in FIG. **13**, but with the body **602** of the head slide **600** removed to illustrate the manner by which the latch linkage **614** cooperates with the latch **604**.

(78) FIGS. **15A** and **15B** are detailed illustrations of the latch **604** and the latch linkage **614**. In the illustrated embodiments, the latch **604** includes a catch **620** and a pin **630**. Catch **620** includes a base member **622** and an engagement member **624** in the illustrated embodiments. The catch **620** is mounted to the body **602** of the head slide **600**. In the illustrated embodiments the base member **622** is mounted to a top portion of the body **602** by a pivot **640** that extends in a direction between the top and bottom portions of the first panel **200**. The base member **622** is thereby configured for rotational movement in a direction extending between the first and second opposite sides of the head slide **600** and first panel **200**, as shown for example by line **623**. The engagement member **624** extends downwardly in a direction parallel to an axis extending between the top and bottom portions of the head slide, and to a location adjacent a first end of the body **602** or adjacent to the first stile **214**. In the illustrated embodiments the engagement member **624** extends in a generally perpendicular direction from the base member **622**. The engagement member **624** is thereby configured to move in a direction such as that shown by line **623** between the first and second opposite sides of the head slide **600**. As shown in FIG. **15B**, the pin **630** is mounted to and extends from the latch linkage **614**. Engagement member **624**, which includes a slot **626** in the illustrated embodiments, extends to a position at which it can be engaged and disengaged by the pin **630**.

(79) The pin **630**, which is driven by the latch linkage **614** in response to actuation of the slide handle **612**, cooperates with the slot **626** to operate or move the latch **604**, and actuate the latch between its latched and unlatched positions or configurations. Control over both the slide mode operation and the pivot mode operation of the first panel **200** are effectively controlled by the operation of the latch **604**. In connection with the pivot mode operation control, when in the unlatched position shown in FIGS. **12-14**, **15A** and **15B**, for example when the slide handle **612** is moved downwardly, the pin **630** is located below the engagement member **624** and the opening of the slot **626**, disengaged from the catch **620**. The first end portion of the first panel **200** is thereby released from head slide **600**, facilitating operation of the first panel in its pivot mode. When the slide handle **612** is moved upwardly, the pin **630** is moved upwardly by the latch linkage **614** and into the slot **626** of the catch **620** to place the latch **604** in the latched position. The first end portion of the first panel **200** is thereby coupled to the head slide **600**, inhibiting pivoting motion of the first panel and facilitating the slide mode operation of the first panel.

(80) In connection with the slide mode operation control, the illustrated embodiments of the fenestration system **10** include a stop structure that inhibits the operation of the latch **604**, e.g. the ability of the latch **604** to be unlatched, unless the first panel **200** is at its pivot ready position. The illustrated embodiments of the stop structure include a stop tab **625** on the catch **620** that cooperates with a stop recess **627** in the guide track **50** (shown in FIG. **16**). The stop tab **625** extends from the catch **620**, and in the illustrated embodiments extends from the base member **622**. The catch **620** is pivotally coupled to the body **602** of the head slide **600** by the pivot **640**. A biasing member such as spring **642** biases the catch **620** toward a stop position, which in the illustrated embodiments is a direction urging the catch to rotate about the pivot **640** away from the head slide **600** to a position where the stop tab **625** extends outwardly from the second side of the head slide toward the guide track **50**. The slot **626** in the engagement member **624** includes portions that slope in a direction from the second side of the first panel **200** (e.g., the right side in FIG. **15A**) toward the first side of the first panel (e.g., the left side in FIG. **15A**) with increasing distance from the base member **622**. Because of the slope of the slot **626** in the catch **620**, the movement of the pin **630** into the slot when the latch **604** is being latched causes the catch to retract against the bias of the spring **642** to a retracted position.

(81) The stop recess **627** is located in the guide track **50** at a position corresponding to the position of the stop tab **625** of the catch **620** when the first panel **200** is located at its pivot ready position. When the first panel **200** is at its pivot ready position and the slide handle **612** is actuated to move the latch to its unlatched position, the stop tab will enter the stop recess **627**. However, if the first panel **200** is at other slide open positions, including slide open positions between the fully open position and the closed position, the stop tab **625** will engage the guide track **50** in response to attempts to unlatch the latch **604**, thereby inhibiting movement of the catch **620** to its stop position and preventing the pin **630** from disengaging from the catch. The head slide **600** is thereby effectively locked into position with respect to the head **112** of the frame **100** when the first panel is located at the pivot ready position. Operation of the latch actuator **610** to configure the first panel **200** for pivot mode operation is also inhibited unless the first panel is at the pivot ready position. Other embodiments of the fenestration system **10** include other or additional structures (not shown) to provide the function of the stop structure described herein. For example, although the stop structure of the embodiments described herein operates effectively simultaneously with the latching functionality of the latch **604**, in other embodiments the stop structure may be configured to operate separately from the latch.

(82) FIGS. **17A** and **17B** illustrate portions of the first and second sides of the fenestration system **10** including a screen carrier to support the screen **400** for operation in both its sliding mode and pivot mode. As described above, when in the sliding mode the screen **400** can slide between a slide closed position, a range of slide open positions, and a pivot ready position. When the screen **400** is in its slide closed position, it may extend across the first lateral portion of the frame **100**

corresponding to the location of the first panel **200** when the first panel is in its slide closed position. When the screen **400** is in any of its slide open positions, the screen at least partially overlaps the second panel **300** when the second panel is in its closed position. When the screen **400** is in its pivot ready position (the position shown in FIGS. **17A** and **17B**), it extends over the second lateral portion of the frame **100** corresponding to the location of the second panel **300** when the second panel is in its closed position, and is engaged with or coupled to the second panel **300** and disengaged from the frame **100**, thereby allowing the screen to move with the second panel when the second panel pivots or hinges between its open and closed positions. Frame **100**, screen **400** and top portions of the second panel **300** are shown in FIGS. **17A** and **17B**, but other portions of the fenestration unit **10** are not shown for purposes of illustration of the screen carrier.

(83) The illustrated embodiments of the screen carrier include a first carrier section **700** mounted to the frame **100**, and a second carrier section **702** mounted to the second panel **300**. In the illustrated embodiments, the first carrier section **700** is mounted to the sill **110** of the frame **100**.

Embodiments of the first carrier section **700** are shown in greater detail in FIG. **18**, where the first carrier section **700** is shown extending over both the first lateral portion and the second lateral portion of the frame **100**, and includes an elongated guide track **704** which is shown for example as a raised rail. In embodiments, the guide track **704** may be a member mounted to the sill **110**. In other embodiments the sill **110** may be a molded or extruded member including the guide track **704** integrally formed therein.

(84) Referring back to FIGS. **17A** and **17B**, the second carrier section **702** is shown mounted to the top rail **312** of the second panel **300** in the illustrated embodiments. Embodiments of the second carrier section **702** are shown in greater detail in FIGS. **19A-19C**, where the second carrier section **702** is shown extending over the second panel **300** between the first stile **314** and the second stile **316**. The second carrier section **702** includes an elongated channel-type guide track **706** in the illustrated embodiments.

(85) FIGS. **20A-20C** illustrate top portions of the frame **100** and second panel **300**. Top portions of the first jamb **114** and the screen **400** are also shown in FIG. **20A**, but not in FIGS. **20B** and **20C**. As shown in FIGS. **17B** and **20A-20C**, embodiments of the screen carrier also include a third carrier section **703** that is mounted to the head **112** of the frame **100**, and that extends over at least portions of the first lateral portion of the frame. The third carrier section **703** extends between the first jamb **114** of the frame **100**, and a first end of the second carrier section **702** in the illustrated embodiments. The third carrier section **703** includes an elongated channel-type guide track **708** in the illustrated embodiments. When the second panel **300** is in its closed position, the first end of the second carrier section **702** and a second end of the third carrier section **703** are located adjacent to one another to facilitate the transition of the screen **400** between the second and third carrier sections during the slide mode operation of the screen. In the illustrated embodiments the second and third carrier sections **702** and **703** are colinear when the second panel **300** is in its closed position. When the second panel **300** is in its closed position, the channels defined by the guide tracks **706** and **708** are sufficiently aligned to facilitate transition of the screen **400** between the second and third carrier sections **702** and **703** during the slide mode operation of the screen. In the illustrated embodiments, the second carrier section **702** has a length that extends over the second panel **300**, and over a portion of the first lateral portion of the frame **100**. The third carrier section **703** extends over only a portion of the first lateral portion of the frame **100** in these embodiments. In other embodiments, the second carrier section **702** and third carrier section **703** have other lengths. For example, in such other embodiments the second carrier section **702** may extend over all or portions of the head **312** of the third panel **300**, and the third carrier section **703** may extend over all or portions of the first lateral portion of the frame **100**.

(86) FIGS. **21A** and **21B** illustrate the first and second ends of the bottom portion of the screen **400**. As shown, the screen **400** includes a plurality of riders **720** (two are shown for purposes of example) that are configured to engage and slide on the sill **110** of the frame **100** while being

guided by the first carrier section **700**. In the illustrated embodiments, the riders **720** include wheels **722** that are rotatably mounted to the bottom rail **410** of the screen **400**, and positioned to engage and roll on the guide track **704** during the slide mode operation of the screen **400**. Riders **720** can include additional or alternative structures in other embodiments (not shown).

(87) FIGS. **22A** and **22B** illustrate the first and second sides of the top portion of the screen **400**.

The illustrated embodiments of screen **400** include two riders **750** that are configured to engage and slide with respect to the head **112** of the frame **100**, while being guided by the second carrier section **702** and the third carrier section **703**. In the illustrated embodiments, each of the riders **750** is an assembly that includes a wheel **752** rotatably mounted to a frame **754**. The riders **750** are pivotally mounted to the top rail **412** of the screen **400** in embodiments. In the illustrated embodiments, for example, a bracket **760** is mounted to the top portion of the screen **400** (e.g., the top rail **412** and/or top portions of the stiles **414** and **416**), and the frames **754** of the riders **750** are pivotally mounted to the bracket **760** by pivots **762**. A spring **764** associated with each of the riders **750** extends between the bracket **760** to an end of the frame **754** opposite the pivot **762** from the wheel **752**. The springs **764** bias the riders **750** in a direction that urges the wheels **752** upwardly (e.g., in a direction away from the top portion of the screen **400**) into engagement with the second and third carrier sections **702** and **703**. When the screen **400** is mounted to the first, second and third carrier sections **700**, **702** and **703**, the riders **750** are therefore biased in a directions that forces the screen downwardly toward the first carrier section **700** to help keep the riders **720** on the bottom portion of the screen engaged with the guide track **704** of the first carrier section **700**.

(88) FIG. **23A** illustrates the screen **400** in its pivot ready position engaged with the second panel **300**, showing the rider **750** adjacent the first stile **414** supported in the guide track **706** of the second carrier section **702**. Although not visible in FIG. **23A**, the rider **750** adjacent the second stile **416** of the screen **400** is similarly supported in the guide track **706** of the second carrier section **702**. FIG. **23B** illustrates the bottom portion of the screen **400** in its pivot ready position. The riders **720**, including the wheels **722**, are disengaged from the sill **110** and guide track **704** when the screen **400** is in the pivot ready position shown in FIGS. **23A** and **23B**.

(89) During the slide mode operation of the screen **400** at and between its slide closed position and the slide open positions (e.g., before the screen is slid to the pivot ready position shown in FIGS. **17A**, **17B**, **23A** and **23B**), the riders **720** on the bottom portion of the screen, including the wheels **722**, are engaged with the first carrier section **700** and guide track **704** on the sill **110**. The bottom portion of the screen **400** is therefore supported and guided by the first carrier section **700** during its slide mode operation at and between its slide closed position and its slide open positions. Similarly, during the slide mode operation of the screen **400** at and between its slide closed position and its slide open positions, the riders **750** on the top portion of the screen, including the wheels **752**, are engaged with the second and/or third carrier sections **702** and **703**. The riders **750** transition between the second carrier section **702** and the third carrier section **703** as the screen **400** slides between its slide open positions and its slide closed positions during the slide mode operation. In embodiments, both of the riders **750** on the top portion of the screen **400** may be engaged with the third carrier section **703**, and thereby slidably engaged with the head **112** of the frame **100**, when the screen **400** is at its closed position. When the screen **400** is at its pivot ready position, both of the riders **750** on the top portion of the screen **400** may be engaged with the second carrier section **702**.

(90) The screen carrier includes one or more disengagement structures that cause the bottom portion of the screen **400** to disengage from the first carrier section **700** and the sill **110** of the frame **100** when the screen **400** is in its pivot ready position. FIGS. **24A** and **24B** illustrate the first and second sides of a disengagement structure **770** that cooperates with the associated rider **750** on the second end of the top portion of the screen **400** (e.g., the side adjacent the second stile **416**) to lift the second end of the screen **400** when the screen transitions to the pivot ready position during the slide mode operation, and to thereby lift and disengage the rider **720** on the second end of the

bottom portion of the screen from the first carrier portion **700**. As shown, the disengagement structure **770** includes a bracket **772** that defines a sloping lift or cam surface **774** on its second side. The associated rider **750** includes a lift or cam follower **776** extending from the second side of the rider **750** to cooperate with the disengagement structure **770** in the illustrated embodiments. The bracket **772** is mounted with respect to or in the second carrier section **702** (not shown in FIGS. **24A-24B**), at locations where the cam follower **776** will engage the cam surface **774** as the screen **400** approaches the end of its range of travel and its pivot ready position during its slide mode operation. The cam surface **774** slopes upwardly and in a direction away from the sill **110** of the frame **100**, with increasing distance toward the second jamb **116** of the frame **100**, and includes a recess or detent **778** at its end. As the screen **400** slides toward its pivot ready position, the cam follower **776** on the associated rider **750** engages and slides on the cam surface **774** until the cam follower engages the detent **778**. The cam follower **776** will be located in the detent **778** when the screen **400** is in its pivot ready position. As the associated rider **750** slides along the cam surface **774** of the bracket **772**, its wheel **752** will continue to engage the second carrier section **702** (not shown in FIGS. **24A** and **24B**), and the frame **754** will rotate about the pivot **762** against the bias force of the spring **764**. Although disengagement structure **770** includes a cam surface **774** on the bracket **772** in the illustrated embodiments, other embodiments include other or additional structures to provide the functionality of the disengagement structure, such as for example a cam surface incorporated into a side wall on the second side of the second carrier section **702**.

(91) FIG. **25** illustrates a disengagement structure **790** that cooperates with the associated rider **750** on the first end (e.g., the end adjacent to the first stile **414**) and top portion of the screen **400** (shown in FIGS. **22A** and **22B**) to lift the first end of the screen **400** when the screen transitions to the pivot ready position during the slide mode operation, and to thereby lift and disengage the rider **720** on the first end of the bottom portion of the screen (shown in FIGS. **21A** and **21B**) from the first carrier portion **700** and sill **110** of the frame **100**. As shown, the illustrated embodiments of the disengagement structure **790** are formed in a first side wall of the second carrier section **702**, and includes a slot **792** that defines a sloping lift or cam surface **794**. The associated rider **750** includes a lift or cam follower **796** (e.g., as shown in FIG. **22A**) extending from the first side of the rider **750** to cooperate with the disengagement structure **790** in the illustrated embodiments. Although shown as being incorporated into the second carrier section **702** in the illustrated embodiments, the disengagement structure **790** may be formed as a separate structure, such as for example the disengagement structure **770** that is mounted to the second carrier section. The disengagement structure **790** is positioned at locations where the cam follower **796** of the associated rider **750** will engage the cam surface **794** as the screen **400** approaches the end of its range of travel and its pivot ready position during its slide mode operation. The cam surface **794** slopes upwardly and away, in a direction away from the sill **110** of the frame **100**, with increasing distance toward the second end of the screen **400**. As the screen slides toward its pivot ready position, the cam follower **796** of the associated rider **750** will engage and slide on the cam surface **794**. As the associated rider **750** slides along the cam surface **794**, its wheel **752** will continue to engage the second carrier section **702**, and the frame **754** will rotate about the pivot **762** against the bias force of the spring **764**. In embodiments, the second carrier section **702** has a length that extends beyond the first stile **314** of the second panel **300** to facilitate a relatively gradual slope of the cam surface **794** and a smooth transition of the screen **400** between its side mode and pivot ready positions.

(92) FIG. **23B** illustrates one or more magnetic structures **755** on the bottom portion of the screen **400** (two are shown on the bottom rail **410** in the illustrated embodiments) that cooperate with other complimentary magnetic structures (not shown) on the bottom portion of the second panel **300** to releasably retain the bottom portion of the screen on the second panel when the screen is at its pivot ready position (e.g., when the riders **720** are disengaged from the sill **110** of the frame **100**). The magnetic structures **755** thereby hold the bottom portion of the screen **400** against the second panel **300** when the second panel, and the screen attached thereto, are moved to their pivot open

positions. Other embodiments (not shown) include additional or alternative, or no, structures to releasably retain the bottom portion of the screen **400** to the second panel **300** during the pivot mode operation of the second panel and screen.

(93) Various modifications and additions can be made to the exemplary embodiments described herein without departing from the spirit and scope of this disclosure. For example, while the embodiments described above refer to particular features, the scope of this disclosure also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of this disclosure is intended to embrace all such alternative, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

Claims

1. A fenestration unit, comprising: a frame including a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb; a first panel supported by the frame; and a slide assembly coupling the first panel to the frame and to facilitate slide mode operation and pivot mode operation of the first panel, wherein during the slide mode operation the first panel can slide within the frame between the first and second jambs when the first panel is in a slide position generally parallel to the frame plane, and during the pivot mode operation the first panel can pivot outwardly beyond the frame plane, and wherein the slide assembly is configured to slide the first panel between a slide closed position, a slide open position, and a pivot ready position beyond the slide open position from the slide closed position, the slide assembly including: a slide body configured for sliding motion between the first and second jambs during the slide mode operation; a pivot rotatably coupling a second end portion of the first panel to a second end portion of the slide body; and a latch releasably coupling the first end portion of the first panel to a first end portion of the slide body, and operable between a latched position and an unlatched position, wherein when the latch is in the latched position the latch couples the first panel to the slide body to facilitate the slide mode operation of the first panel, and wherein when the latch is in the unlatched position the latch releases the first panel from the slide body to facilitate the pivot mode operation of the first panel, and wherein the latch includes a stop that inhibits actuation of the latch to the unlatched position unless the first panel is at the pivot ready position; and a latch actuator coupled to the latch and operable by a user to cause the latch to move between the latched position and the unlatched position.

2. The fenestration unit of claim 1, wherein the latch actuator is located on the first panel.

3. The fenestration unit of claim 1, wherein: the frame includes a slide assembly guide track, and wherein the slide body of the slide assembly is slidably mounted in the slide assembly guide track; and the slide assembly guide track includes a structure configured to cooperate with the stop of the latch when the first panel is at the pivot ready position to enable the actuation of the latch to the unlatched position.

4. The fenestration unit of claim 3, wherein the guide track includes a structure configured to cooperate with the stop of the latch when the first panel is at and between its slide closed position and its slide open position, to inhibit the actuation of the latch to the unlatched position.

5. The fenestration unit of claim 1, wherein the frame includes a slide assembly guide track, and wherein the slide body of the slide assembly is slidably mounted in the slide assembly guide track.

6. The fenestration unit of claim 5, wherein: the slide assembly guide track is on the head of the frame, and the slide assembly couples a top portion of the first panel to the head of the frame.

7. The fenestration unit of claim 1, wherein: the first panel is defined by first and second opposite ends; and the slide body extends an entire distance between the first and second opposite ends of the first panel.

8. The fenestration unit of claim 1, wherein the latch comprises: a catch movably mounted to the

first panel; and a pin that is actuated by the latch actuator to releasably engage the catch.

9. The fenestration unit of claim 8, wherein movement of the pin causes the latch to move between the latched position and the unlatched position.

10. The fenestration unit of claim 9, wherein the catch comprises: a base member; and an engagement member extending from the base member, wherein the engagement member includes a slot that can be engaged and disengaged by the pin.

11. The fenestration unit of claim 10, wherein: the base member is pivotally mounted to the first panel for rotational movement about a pivot axis extending in a direction between a top portion and a bottom portion of the first panel; and the engagement member extends in a generally perpendicular direction from the base member.

12. The fenestration unit of claim 11, wherein: the base member is pivotally mounted to one of a top portion or a bottom portion of the first panel; and the engagement member extends in a direction generally parallel to an end edge of the first panel.

13. The fenestration unit of claim 12, wherein the stop is on the base member.

14. The fenestration unit of claim 13, wherein: the frame includes a slide assembly guide track, and wherein the slide body of the slide assembly is slidably mounted in the slide assembly guide track; and the slide assembly guide track includes a structure configured to cooperate with the stop on the base member when the first panel is at the pivot ready position to enable the actuation of the latch to the unlatched position.

15. The fenestration unit of claim 14, wherein the guide track includes a structure configured to cooperate with the stop on the base member when the first panel is at and between its slide closed position and its slide open position, to inhibit the actuation of the latch to the unlatched position.

16. The fenestration unit of claim 14, wherein the slot of the engagement member slopes to inhibit the actuation of the latch to the unlatched position when the first panel is at and between its slide closed and slide open positions.

17. The fenestration unit of claim 1, further comprising: a second panel supported by the frame to facilitate hinge mode operation of the second panel, wherein during the hinge mode operation the second panel can pivot between a closed position generally parallel to the frame plane and an open position beyond the frame plane; and one or more stop members configured to prevent the first panel from sliding from the slide open position to the pivot ready position during the slide mode operation when the second panel is in its closed position.

18. A fenestration unit, comprising: a frame defining a frame plane and including a sill, a head opposite the sill, a first jamb and a second jamb opposite the first jamb; a first panel supported by the frame; and a locking pivot rider assembly coupling one of a top or bottom portion of the first panel to the frame and to facilitate slide mode operation and pivot mode operation of the first panel, wherein during the slide mode operation the first panel is in a slide position generally parallel to the frame plane and can slide within the frame between the first and second jambs, and during the pivot mode operation the first panel can pivot outwardly to a pivot open position beyond the frame plane, the locking pivot rider assembly including: a rider coupling the first panel to the frame for the slide mode operation; a pivot rotatably coupling the first panel to the rider for the pivot mode operation; and a lock coupled to the rider and the pivot, to releasably lock the rider with respect to the frame, wherein rotation of the pivot during the pivot mode operation of the first panel from the slide position to the pivot open position actuates the lock and causes the lock to inhibit sliding movement of the rider on the frame, and wherein rotation of the pivot during the pivot mode operation of the first panel from the pivot open position to the slide position releases the lock to allow sliding movement of the rider on the frame; and a slide assembly coupling the other of the top or bottom of the first panel to the frame and to facilitate the slide mode operation and the pivot mode operation of the first panel, the slide assembly including: a slide body configured for sliding motion between the first and second jambs during the slide mode operation; a pivot rotatably coupling a second end portion of the first panel to a second end portion of the slide body; and a

latch releasably coupling a first end portion of the first panel to a first end portion of the slide body, and operable between a latched position and an unlatched position, wherein when the latch is in the latched position the latch couples the first panel to the slide body to facilitate the slide mode operation of the first panel, and wherein when the latch is in the unlatched position the latch releases the first panel from the slide body to facilitate the pivot mode operation of the first panel; and a latch actuator coupled to the latch and operable by a user to cause the latch to move between the latched position and the unlatched position.

19. The fenestration unit of claim 18, wherein: the slide assembly is configured to slide the first panel between a slide closed position, a slide open position, and a pivot ready position beyond the slide open position from the slide closed position; and the latch further includes a stop that inhibits actuation of the latch to the unlatched position unless the first panel is at the pivot ready position.

20. The fenestration unit of claim 19, wherein: the frame includes a slide assembly guide track, and wherein the slide body of the slide assembly is slidably mounted in the slide assembly guide track; and the slide assembly guide track includes a structure configured to cooperate with the stop of the latch when the first panel is at the pivot ready position to enable the actuation of the latch to the unlatched position.

21. The fenestration unit of claim 20, wherein the guide track includes a structure configured to cooperate with the stop of the latch when the first panel is at and between its slide closed position and its slide open position, to inhibit the actuation of the latch to the unlatched position.

22. The fenestration unit of claim 19, wherein: when the first panel is in the slide closed position and the second panel is in the closed position, the first panel and the second panel are parallel with one another, and when the first panel is in the slide open position and the second panel is in the closed position, the first and second panels are parallel with one another and the first panel is at a lateral position that at least partially overlaps the second panel; and the one or more stop members comprises a bumper coupled to the second panel.
